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The effect of cervical spine flexion-extension motion on odontoid parameters

Longao Huang^{1†}, Dun Liu^{1†}, Hongyuan Xu^{1†}, Junfei Feng¹, Tao Kang¹, Shengwang Wei² and Hua Jiang^{1*}

Abstract

Objective To assess the stability of odontoid parameters on flexion-extension motion and to validate the accuracy of the physiological cervical lordosis (CL) predictive formula across different cervical positions.

Methods Standard cervical spine lateral radiographs in neutral, flexion, and extension positions were collected to measure odontoid incidence (OI), odontoid tilt (OT), C2 slope (C2S), CL, T1 slope (T1S), and T1S minus CL (T1S-CL). Friedman's test was used to assess the differences in parameters among the three cervical spine positions. The predictive performance of the formula $CL = 0.36 \times OI - 0.67 \times OT - 0.69 \times T1S$ was assessed and validated using the Pearson correlation coefficient (r), coefficient of determination (R^2), mean squared error (MSE), and mean absolute error (MAE).

Results No significant differences were found for OI and T1S among the three different cervical spine positions ($p = 0.162$ and $p = 0.186$, respectively). There was a strong, significant positive correlation between the predicted value and actual value of physiological CL at three cervical positions (neutral, flexion, and extension). The predictions for the neutral position were the most accurate, with statistical measures of $r = 0.85$ ($p < 0.01$), $R^2 = 0.82$, $MAE = 4.28$, and $MSE = 27.77$.

Conclusion OI is a stable and reliable anatomic parameter, not affected by cervical spine flexion-extension motion. OI can serve as a supplementary parameter for evaluating cervical sagittal balance and compensatory ability. The formula $CL = 0.36 \times OI - 0.67 \times OT - 0.69 \times T1S$ provides the best predictions for physiological CL in the neutral position.

Keywords Cervical flexion and extension, Cervical spine motion, CL prediction, Odontoid incidence

Introduction

Cervical degenerative disease is a major cause of spinal diseases, and its incidence has been rising in the elderly people [1]. Evaluations of spinal sagittal balance frequently inform surgical decision-making in cervical degenerative disease [2]. C2 Slope (C2S), C2-7 angle (cervical lordosis, CL), T1 Slope (T1S), and T1S minus CL (T1S-CL) are good cervical parameters to evaluate cervical sagittal balance [3–6]. However, these positional parameters are flawed because they are not constant anatomical parameters, and the measured values may have significant errors [7].

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Inspired by the pelvic parameters, Lee et al. [8, 9] proposed a novel concept of odontoid parameters as an adjunct to the positional parameters above to overcome this flaw. Recent studies have shown that the odontoid parameters were significantly correlated with established cervical parameters and health-related quality of life (HRQOL) measures following multilevel posterior cervical fusion [9]. Odontoid incidence (OI) is a crucial anatomical parameter for evaluating cervical sagittal balance, the algebraic sum of odontoid tilt (OT) and C2S [8]. Pelvic incidence (PI) is an individualized morphological parameter, not affected by the individual posture or position, and is considered to be a permanently invariable parameter for a subject [10, 11]. After skeletal maturity, this angle is fixed [12]. As the PI, the OI is the only anatomical and constant parameter that could be used as a reference point related to the cervical spine from the rostral end [8]. The measurement of OI incorporates both horizontal and vertical axes, enabling the analysis of cervical spine alignment in two dimensions [8]. In addition, OI represents the compensatory reserve or range of extension of the upper cervical spine in individuals [9]. Studies have shown that patients with a larger OI experience more significant postoperative improvements in HRQOL [9]. Additionally, OI is closely related to the range of motion and extension angles of C0-2 and C1-2, highlighting the importance of OI in evaluating cervical function and compensatory ability [9]. OI is correlated with various cervical sagittal parameters in asymptomatic individuals, adolescent idiopathic scoliosis (AIS) patients, and cervical spondylotic myelopathy (CSM) patients, further demonstrating its clinical potential in assessing cervical compensatory status [8, 13, 14]. Although OI shows promise as a tool for evaluating cervical sagittal balance, its clinical application may differ significantly from PI. PI can be directly used to guide surgical correction angles [10], whereas OI is more inclined to serve as a supplementary indicator for assessing cervical alignment and compensatory ability. In multilevel posterior cervical fusion surgery, OI is significantly associated with postoperative HRQOL improvement. This suggests that OI can be used as an auxiliary tool to provide reference for individualized surgical planning, optimize corrective strategies, and reduce postoperative complications. However, OI has not yet reached the widespread clinical use of PI, and further research is needed to validate its applicability in different surgical approaches and pathological conditions.

Although preliminary studies have confirmed that OI is a constant anatomical parameter, current research mainly focuses on static analysis or neutral position measurements, lacking exploration of OI stability during dynamic cervical movements [8, 9, 14, 15]. Cervical flexion and extension movements may cause changes in

vertebral angles, intervertebral discs, and sagittal alignment. Therefore, verifying the stability of OI in dynamic states can further confirm its potential as a reliable and stable parameter for evaluating cervical sagittal alignment. On the other hand, the dynamic changes in the cervical vertebrae across different postures and their effects on OI, along with the reciprocal relationships between changes in OI, OT, and C2S, have been underinvestigated. Similarly, the predictive formula $CL = 0.36 \times OI - 0.67 \times OT - 0.69 \times T1S$ proposed by Lee et al. [8] to predict physiological CL in asymptomatic populations remains to be verified for accuracy with changes in cervical position.

This study aimed to evaluate the stability of OI during flexion-extension motion of the cervical spine in standing asymptomatic populations. In addition, it analyzed changes in odontoid parameters, including OI, OT, and C2S, as well as their associated parameters, including T1S, CL, and T1S-CL, in neutral, flexion, and extension positions. Finally, this study will validate the physiological CL predictive formula against changes in cervical position. Analyzing how these parameters perform in asymptomatic individuals provides a reference for the evaluation of sagittal balance and the development of surgical strategies in clinical practice. This may assist surgeons in more precisely assessing and correcting cervical deformities.

Materials and methods

Study population

This study retrospectively collected imaging data of healthy volunteers who underwent cervical health assessments at our hospital from January 1, 2022, to December 31, 2023. The study subjects were individuals who have been engaged in office work for a long time, as this occupation may significantly impact cervical health. To evaluate the effect of office work on cervical function, we added hyperflexion and hyperextension imaging examinations to the routine cervical examinations.

Each subject was informed of the procedures, methods, benefits, and possible risks of the study. Informed consent for imaging examinations and data usage was obtained from all individual participants included in the study. The study protocol was approved by our Institutional Review Board (approval number:2024-E083-01). The research complies with the Declaration of Helsinki.

Inclusion and exclusion criteria

The inclusion criteria were as follows: (1) The age was 18–48 years old; (2) no history of symptoms, diagnosis, and treatment related to the whole spine, hip joint, and lower extremity. No history of chronic pain in the neck or shoulder, no history of spinal diseases or surgery, no history of pelvis, hip joint, or lower extremity diseases;

(3) no cervical instability and spondylolisthesis showed on X-ray; (4) no significant scoliosis in the coronal plane (Cobb angle $< 10^\circ$), and no kyphosis deformity in the sagittal plane; (5) standard standing cervical spine lateral X-rays, including neutral, flexion, and extension positions. A total of 98 volunteers were enrolled, including 56 males and 42 females.

To improve the validity and reliability of the study, the following screening criteria were adopted to ensure participants' health status and exclude possible clinical interference factors. Volunteers were excluded from the study if they fulfilled any of the following criteria: (1) Cobb angle of 10° or greater in the coronal plane; (2) the cervical morphology conforms to any one of the Straight, Sigmoid, and Kyphotic types in accordance with the modified method proposed by Toyama et al. [16, 17]; (3) a history of major surgeries of the spine, hip, or knees such as joint replacement, ligament reconstruction, fracture fixation, or spinal fusion; (4) a history of neuromuscular disorders or inflammatory arthritis; (5) recent back pain, neck pain, or lower extremity pain that influenced activities of daily living resulting in dependence on narcotics; (6) inability to stand without assistance; and (7) pregnancy.

We implemented strict inclusion and exclusion criteria to ensure a balance between homogeneity and robustness, aiming to minimize potential bias. Finally, the study included 50 volunteers.

Acquisition conditions

All the radiographs were taken using the same method and under the same conditions. A standard radiographic examination was required for all volunteers: with uniform position of an upright stance with relaxed shoulders, maintaining leveled vision with both eyes looking straight ahead, avoiding "soldier posture", and other details including lowered chins that affected stance posture [18, 19]. With the central line pinpointed toward C4

and the projection distance of 1.5 m, we obtained standard right sagittal views, with the upper boundary of the radiograph film parallel to the hard palate, requiring maximum flexion and extension bending without additional external force [17]. We previewed our films to assess standardization with viable results through preliminary radiographic scanning before actual projection; we selected automatic exposure parameters, and one technician conducted and completed all radiophotography to ensure uniformity.

The radiographic imaging device used in this study was the AXIOM-Artis VX (Siemens, Erlangen, Germany), employed for X-ray imaging. The radiographs were recorded on a PACS/RIS system (Neusoft Hospital Information System, HIS 5.0, China). All images were stored in DICOM format and then transferred to personal computers for analysis. In this study, concerning the C2–7 angle, positive and negative values indicated lordosis and kyphosis, respectively [20].

Data collection and measurement

Data were collected by three investigators in the research team, who would not be involved in the final data analysis. Two certified senior spine surgeons (with 12 and 6 years' clinical experience, respectively) used the Surgimap software (Nemaris, Inc., New York, NY, USA) to measure the following parameters on all radiographs (Fig. 1). OI was defined as the angle between the line perpendicular to the C2 endplate (C2EP) at its midpoint and the line connecting this point to the center of the odontoid process (the center of a circle with an anterior/posterior border and the apex of the dens as a tangent) (Fig. 1) [9]. OT was defined as the angle created by a line running from the C2EP midpoint to the center of the odontoid process and the vertical axis [8]. Negative values indicated that the center of the odontoid process was placed anterior to the C2EP midpoint [9]. C2S was defined as the angle between the C2EP and a horizontal line [8]. T1S is

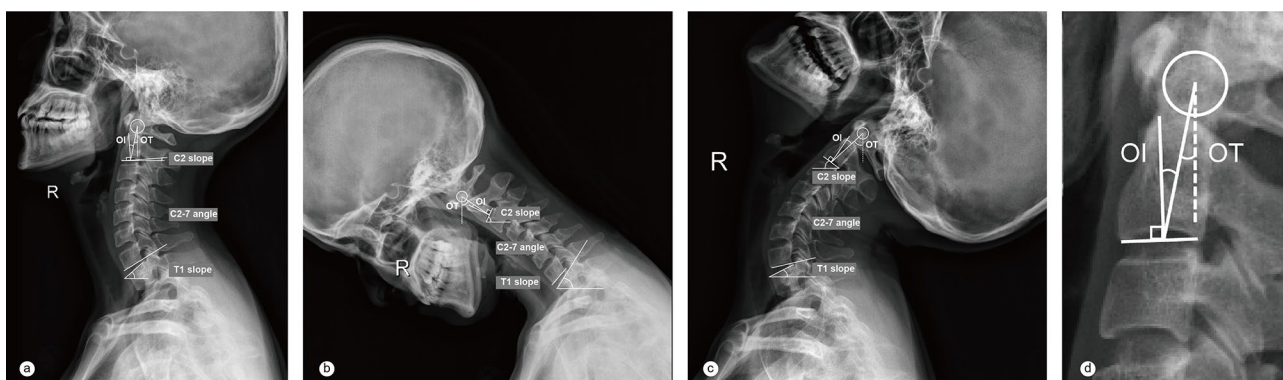


Fig. 1 Measurement of odontoid and cervical spine parameters on lateral cervical radiographs of a healthy 28-year-old female in three different positions: neutral (A), flexion (B), and extension (C), with high-resolution magnified views of OI and OT in the neutral position (D). OI, odontoid incidence; OT, odontoid tilt

the intersection angle between the upper edge of T1 and the horizontal line [9]. The C2-7 angle was defined as the angle of the segment from the C2EP to the inferior C7EP [21]. T1S-CL: T1S minus CL. All measurements were repeated after a 6-week washout period to ensure the accuracy of the results. All basic radiographic parameters were each measured twice by two senior spine surgeons, and the mean values were utilized for statistical analysis.

Statistical analyzes

All statistical analyses were performed using SPSS Statistics (version 26, IBM SPSS Inc., USA), R (version 4.2.2), RStudio (version 1.1.463), and GraphPad Prism software (version 9.4.1). In addition, the normality of the data was checked using the Shapiro-Wilk test. Normally distributed data were presented as mean±standard deviation or as median (interquartile range) if they were not normally distributed. Odontoid parameters, along with their associated parameters, in the neck’s neutral, flexion, and extension positions were compared using Friedman’s test. Post-hoc tests with Bonferroni corrections were applied for pairwise comparisons. Comparisons of parameters between flexion and neutral positions, as well as extension and neutral positions, were conducted using paired t-tests or Wilcoxon signed rank tests as appropriate. Bivariate correlations of changes for each parameter between flexion and neutral positions, as well as extension and neutral positions, were obtained through the Pearson or Spearman method. All tests were two-tailed. Statistical significance was set a $p < 0.05$. The strength of the variable correlation was classified based on guidelines suggested by Cohen: ($r = 0.10–0.29$) = weak correlation, ($r = 0.30–0.49$) = moderate correlation, and ($r = 0.50–1.0$) = strong correlation [22].

Results

Study subjects and data collection

A total of 98 individuals who participated in health checkups were screened in this study, of whom 48 were excluded based on the exclusion criteria. The study

population consisted of 50 participants [24 (48%) males and 26 (52%) females] with a mean age of 29.78 ± 7.87 (18–48 years old). The study involved the measurement of six sagittal parameters, acquiring a total of 1800 measurements.

Intra- and inter-observer reliability of cervical spine measurements

Intraclass correlation coefficients (ICCs) were performed using a two-way mixed effects model for intra-observer reliability and a two-way random effects model for inter-observer reliability (absolute agreement, average measures) [19]. This was used to calculate the intra-observer and inter-observer reliability. ICC of 0.75 to 1.00 was indicative of excellent test-retest reliability [23]. Table 1 summarizes the intra-observer and inter-observer reliability for various parameters in the cervical spine’s neutral, flexion, and extension positions. All parameters showed a high reliability with an ICC of up to 0.834.

The influence of cervical spine flexion-extension motion on the cervical sagittal parameters

In this study, we assessed the critical parameters in the cervical spine’s neutral, flexion, and extension positions to evaluate the influence of flexion-extension motion on sagittal cervical parameters (Table 2). The Friedman test, with Dunn’s correction, showed a significant difference ($p < 0.001$) among the three different cervical spine positions for OT, C2S, CL, and T1S-CL, while no significant difference was found for OI and T1S ($p = 0.162$ and $p = 0.186$, respectively). In detail, C2S and CL showed a significant increase when compared with the neutral position in the flexion position, but a significant reduction was observed in the extension position. The change in OT was demonstrated, and it exhibited an opposite trend compared with C2S and CL. Table 3 illustrates the differences between all parameters in the maximum flexion and extension positions compared to the initial neutral position. The subjects were then subclassified into three groups to gain perspective into the global trends

Table 1 Reliability of the odontoid and cervical parameters

Parameters	Neutral			Flexion			Extension		
	Inter-rater		Intra-rater	Inter-rater		Intra-rater	Inter-rater		Intra-rater
	Rater 1–2	Rater 1	Rater 2	Rater 1–2	Rater 1	Rater 2	Rater 1–2	Rater 1	Rater 2
OI	0.871	0.985	0.963	0.834	0.985	0.970	0.882	0.951	0.955
OT	0.968	0.996	0.995	0.975	0.998	0.996	0.952	0.997	0.996
C2S	0.955	0.997	0.997	0.966	0.998	0.974	0.967	0.998	0.996
T1S	0.866	0.964	0.910	0.934	0.968	0.936	0.904	0.983	0.915
CL	0.954	0.994	0.991	0.967	0.991	0.981	0.990	0.999	0.997
T1S-CL	0.983	0.994	0.989	0.954	0.982	0.969	0.980	0.997	0.994

Abbreviations: CI, confidence interval; ICC, intraclass correlation coefficient; C2S, C2 slope; CL, cervical lordosis; OI, odontoid incidence; OT, odontoid tilt; T1S, T1 slope; T1S-CL, T1S minus CL

All values were statistically significant ($p < 0.001$)

Table 2 Angles of the odontoid and cervical parameters in neutral, flexion, and extension positions

Parameters	Neutral	Flexion	Extension	P-value ^a	Subgroup comparison ^a
OI (°)	16.31 ± 3.27	16.4(4.29)	15.33 ± 3.78	0.162	-
OT (°)	14.96 ± 7.72	-40.89 ± 12.57	36.13(16.36)	< 0.001*	b* c* d*
C2S (°)	3.88(11.96)	52.30 ± 12.96	-21.45(16.04)	< 0.001*	b* c* d*
CL (°)	-23.61 ± 9.03	23.58 ± 10.38	-46.55(18.17)	< 0.001*	b* c* d*
T1S (°)	32.47 ± 4.95	33.83 ± 8.85	29.50(9.99)	0.186	-
T1S-CL (°)	56.08 ± 11.87	10.25 ± 14.89	74.10(21.70)	< 0.001*	b* c* d*

Abbreviations: C2S, C2 slope; CL, cervical lordosis; OI, odontoid incidence; OT, odontoid tilt; T1S, T1 slope; T1S-CL, T1S minus CL. *Statistical significance with $p < 0.05$. Normally distributed data were presented as mean ± standard deviation, or as median (interquartile range) if they were not normally distributed

^a Friedman test, followed by post-hoc pairwise comparison

b: Flexion vs. Neutral

c: Extension vs. Neutral

d: Flexion vs. Extension

Table 3 Changes in measurement of the odontoid and cervical parameters with cervical spine flexion-extension motion

Parameters	Statistic	Flexion			Extension		
		Increase	Decrease	No change	Increase	Decrease	No change
OI	N	26	24	0	18	32	0
	Mean (Range)	2°(0.15°, 12.30°)*	1.99°(-6.95°, -0.10°)*		1.3°(0.05°, 4.50°)*	2.28°(-11.6°, -0.1°)*	
OT	N	0	50	0	49	1	0
	Mean (Range)		55.86°(-85.25°, -26.25°)*		21.29°(0.80°, 65.55°)*	41.6°(-41.6°, -41.6°)	
C2S	N	50	0	0	1	49	0
	Mean (Range)	55.94°(27.90°, 83.25°)*			39.35°(39.35°, 39.35°)	22.26°(-70.85°, -3.60°)*	
CL	N	50	0	0	2	48	0
	Mean (Range)	47.19°(19.7°, 75.5°)*			85.13°(67.60°, 102.65°)#	23.01°(-41.35°, -5.50°)*	
T1S	N	26	24	0	22	28	0
	Mean (Range)	6.85°(0.80°, 20.75°)*	4.6°(-11.85°, -0.25°)*		3.77°(0.25°, 13.30°)*	11.5°(-77.3°, -0.6°)*	
T1S-CL	N	0	50	0	48	2	0
	Mean (Range)		45.84°(-77.8°, -16.8°)*		21.08°(0.3°, 53.3°)*	158.4°(-179.95°, -136.85°)#	

Abbreviations: C2S, C2 slope; CL, cervical lordosis; OI, odontoid incidence; OT, odontoid tilt; T1S, T1 slope; T1S-CL, T1S minus CL; N, number of participants. *Statistical significance with $p < 0.05$. #Not significant with $p > 0.05$

Increase: The parameter at the flexion-extension endpoint positions has increased compared to the neutral position

Decrease: The parameter at the flexion-extension endpoint positions has decreased compared to the neutral position

No change: The parameter at the flexion-extension endpoint positions shows no change compared to the neutral position

of sagittal cervical parameters' excursion with cervical position changes. The parameters at the endpoint positions of flexion-extension motion respond with either an increase, decrease or no change when compared to the initial neutral position. Statistically significant differences were found for the changes of all parameters in cervical flexion. In cervical extension, an increase in CL and a decrease in T1S-CL showed no statistical differences ($p = 0.18$), while the changes in all other parameters were significant.

The examination of cervical spine flexion-extension motion on the OI and T1S parameters revealed no clear trend. In cervical flexion, the changes in OI and T1S were evenly distributed among subjects, with 26 and 24

subjects showing increases or decreases, respectively. In contrast, the parameters of OT, C2S, CL, and T1S-CL show an obvious trend of variation during cervical spine flexion-extension motion. In cervical flexion, C2S and CL showed an average increase of 55.94° and 47.19°, respectively, while OT and T1S-CL showed an average decrease of 55.86° and 45.84°. In cervical extension, C2S and CL showed an average decrease of 22.26° and 23.01°, respectively, while OT and T1S-CL showed an average increase of 21.29° and 21.08°.

Table 4 Pairwise comparison of changes in each parameter between flexion and neutral positions

Parameters	OI	OT	C2S	CL	T1S	T1S-CL
OI		0.097	0.063	0.145	-0.127	-0.152
OT			-0.974*	-0.698*	-0.531*	0.232
C2S				0.749*	0.506*	-0.270
CL					-0.094	-0.769*
T1S						0.631*
T1S-CL						

Abbreviations: C2S, C2 slope; CL, cervical lordosis; OI, odontoid incidence; OT, odontoid tilt; T1S, T1 slope; T1S-CL, T1S minus CL

* Significant correlation at the level of $p < 0.01$ **Table 5** Pairwise comparison of changes in each parameter between extension and neutral positions

Parameters	OI	OT	C2S	CL	T1S	T1S-CL
OI		0.020	0.220	0.093	0.231	0.046
OT			-0.954*	-0.745*	-0.053	0.529*
C2S				0.738*	0.137	-0.496*
CL					-0.342†	-0.896*
T1S						0.674*
T1S-CL						

Abbreviations: C2S, C2 slope; CL, cervical lordosis; OI, odontoid incidence; OT, odontoid tilt; T1S, T1 slope; T1S-CL, T1S minus CL

* Significant correlation at the level of $p < 0.01$ † Significant correlation at the level of $p < 0.05$

The analysis of the correlations between the changes of each parameter during cervical spine flexion-extension motion

Tables 4 and 5 illustrate the correlations between changes in each parameter during cervical spine flexion-extension motion. During cervical spine flexion-extension motion, changes in OI showed no correlation with changes in the other parameters. In cervical flexion, significant positive correlations were observed between the changes in C2S and the changes in CL and T1S ($r = 0.749$ and $r = 0.506$, respectively). All p -values were < 0.01 . Significant negative correlations were observed between the changes in OT and the changes in C2S, CL, and T1S ($r = -0.974$, $r = -0.698$, and $r = -0.531$, respectively). All p -values were < 0.01 . In cervical extension, significant positive correlations were observed between the changes of C2S and CL ($r = 0.738$, $p < 0.01$), while significant negative correlations were observed between the changes of C2S and T1S-CL ($r = -0.496$, $p < 0.01$). The changes of OT showed significant negative correlations with the changes of C2S and CL ($r = -0.954$ and $r = -0.745$, respectively, $p < 0.01$), and a significant positive correlation with the changes of T1S-CL ($r = 0.529$, $p < 0.01$).

The statistical performance measures of the predictive formula at different cervical positions

We evaluated the statistical performance measures of the predictive formula $CL = 0.36 \times OI - 0.67 \times OT - 0.69 \times T1S$ at different cervical positions (neutral, flexion, and extension). These measures included the Spearman correlation coefficient, Pearson correlation coefficient, coefficient

of determination (R^2), mean squared error (MSE), and mean absolute error (MAE). R^2 is a number between 0 and 1 that measures how well a statistical model predicts an outcome [24]. MAE and MSE can directly measure the degree of deviation between the actual value and the predicted value [25]. Lower values of MAE and MSE indicate better predictive performance [26]. Figure 2 showed a significant, strong, positive correlation between the predicted value and actual value of physiological CL ($r = 0.85$, $p < 0.01$) in the neutral position. As can be seen from Fig. 2, the MSE and MAE are lowest in the neutral position, at 27.77 and 4.28 respectively, indicating the best prediction performance. The R^2 for the neutral position is 0.82, indicating that the model explains approximately 82% of the variation in CL. By contrast, the MSE and MAE in the flexion position are significantly higher than those of the other positions, at 209.23 and 13.64, respectively. For further refinement of the prediction formula, we analyzed the difference distributions between the actual values and predicted values at different cervical positions (neutral, flexion, and extension).

As shown in Fig. 3, in the neutral position, the differences between the predicted value and actual value were mainly (72%) equal to or less than 5° , with only 8% of the differences between 11 – 15° and none exceeding 15° . In the flexion position, the differences between the predicted value and actual value are notably large, with 38% of the differences exceeding 15° . Taken together, the formulae to predict physiological CL showed the best predictive performance in the neutral position.

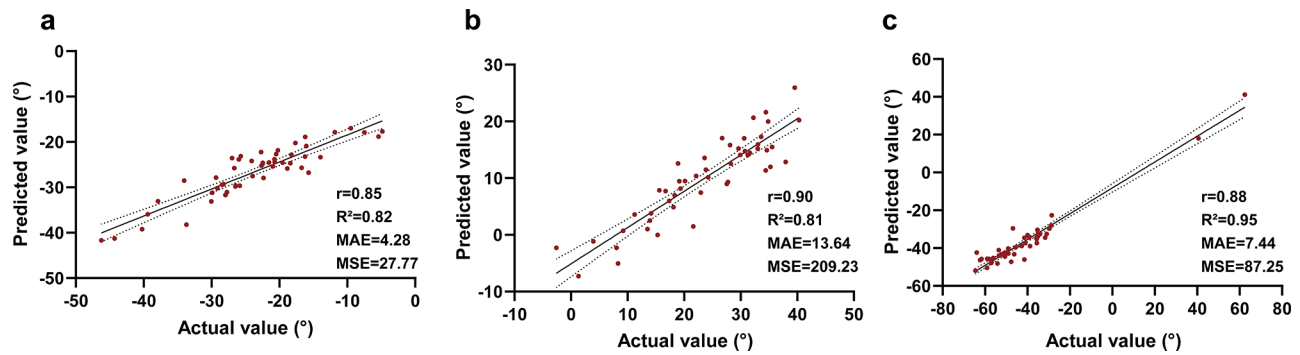


Fig. 2 Validation of the physiological CL predictive formula on lateral cervical radiographs in three different positions: neutral (A), flexion (B), and extension (C). CL, cervical lordosis; r , Pearson or Spearman correlation coefficient; R^2 , coefficient of determination; MAE, mean absolute error; MSE, mean squared error

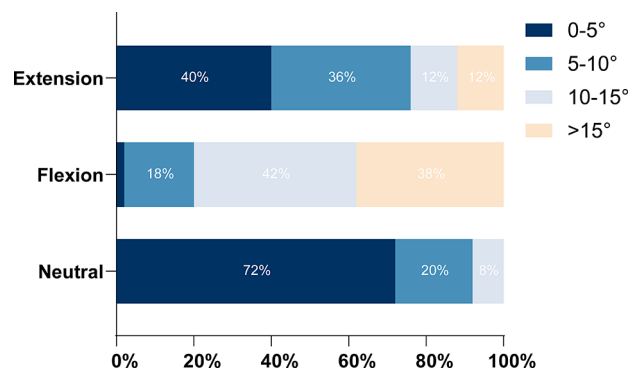


Fig. 3 Distribution of CL prediction error (0–5°, 5–10°, 10–15°, >15°) on lateral cervical radiographs in neutral, flexion, and extension positions

Discussion

Sagittal cervical parameters, including C2S, CL, T1S, and T1S-CL, are associated with HRQOL and have important implications for the selection of surgical methods and the evaluation of postoperative sagittal balance [27–30]. However, these parameters are posture-related and influenced by the position of the cervical spine, leading to uncertainty in the selection of surgical methods and inaccurate sagittal plane assessment [9, 14]. Therefore, discovering new independent and constant anatomic parameters is crucial for an in-depth evaluation of cervical sagittal balance.

This study is the first to find the stability of OI during flexion-extension motion of the cervical spine. Our results show that there were no statistically significant differences in the OI values between the neutral, flexion, and extension positions ($p=0.162$), supporting the reliability of OI as a stable anatomic parameter [8, 14, 15]. Lee et al. [8] found that OI had significant correlations with cervical alignment and is the only anatomical and constant parameter that could be used as a reference point related to the cervical spine from the rostral end. In asymptomatic subjects, OI showed significant correlations with the C0–2 angle, C2–7 angle, and C2S [8]. Other studies have shown that OI is significantly

correlated with patients' HRQOL scores, with those having a larger OI exhibiting better postoperative HRQOL scores [9]. Our previous research indicated that in the healthy asymptomatic population and patients with cervical spondylosis, there were no statistically significant differences between standing plain radiographs and supine MRI in measuring OI [14, 15]. This proves that its measurement was not influenced by external factors such as body position and examination instruments. Previous studies indicate that OI is significantly associated with various cervical sagittal parameters across different populations, including healthy individuals, CSM patients, AIS patients, and Chinese children without cervical symptoms [13–15, 31]. Therefore, OI, as a stable and reliable anatomical parameter, might be a key factor in regulating cervical sagittal alignment. However, the reference value of 17° for OI is based on a small sample size, and its generalizability needs further validation. The threshold of OI related to HRQOL impairment and age-related standards are still unclear, limiting the effectiveness and clinical application of the research results. Future studies need to further validate OI in larger and more homogeneous populations to determine its normal range and dynamic changes, thereby reducing potential clinical biases and enhancing the generalizability of research and practice.

During cervical spine flexion-extension motion, OT, C2S, CL, and T1S-CL all showed significant differences, consistent with previous studies, confirming that these parameters are highly susceptible to body posture [4, 8, 14, 15]. Our results show that in cervical flexion, C2S and CL significantly increase while OT and T1S-CL significantly decrease. Conversely, in cervical extension, C2S and CL significantly decrease while OT and T1S-CL significantly increase. In cervical flexion, the increase in C2S is significantly positively related to the increases in CL and T1S. In addition, there were significant negative correlations between C2S and OT during cervical spine flexion-extension motion. These results are consistent

with recent findings from related studies, which indicated that C2S has an inverse relationship with OT [8]. This also confirmed that there is a mutual compensation mechanism between OT and C2S during cervical spine flexion-extension motion. Specifically, as C2S increases, the vertical axis from the center of the odontoid is placed anterior to C2EP, resulting in a negative OT value [8]. In summary, the dynamic change of C2S during cervical spine flexion-extension motion and its synergistic relationship with other parameters suggest that C2S could be used in assessing global cervical deformities.

Pelvic tilt (PT) is a posture-related parameter used to define the amount of pelvic compensation [32]. It indicates the spatial orientation of the pelvis and reflects the pelvic flexion or posterior tilt angle [33]. Increased PT during the standing position, which reflects the compensation for proximal spine deformity, is considered one of the most important parameters affecting HRQOL scores [34, 35]. OT, like PT, denotes the spatial orientation of the dens, which may vary according to the balance of the cranium and horizontal gaze [9, 32]. OT was found to be correlated with the NDI score ($r=-0.37$, $p<0.05$) in the studies of Lee et al. [9]. In our previous study, we found significant relationships between OT, C2S, CL, T1S, and T1S-CL [15]. These relationships together revealed the connection between the structural geometry of the odontoid process and the sagittal alignment of the cervical spine. In this study, in cervical flexion, the decrease in OT is significantly negatively related to the increases in C2S, CL, and T1S. In cervical extension, the increase in OT is significantly negatively related to the decreases in C2S and CL. Based on these findings, we speculated that the value of OT might somehow be related to cervical deformity. While C2S is a simplified measure, OT offers a more detailed assessment of the odontoid's orientation, which could provide nuanced insights into cervical alignment and its impact on the cranium and visual gaze [8, 9]. This level of detail might be crucial in certain clinical scenarios where fine adjustments and precision are essential. However, the dynamic changes of OT and C2S during cervical flexion and extension may introduce measurement bias and uncertainty. These changes can lead to fluctuations in HRQOL-related parameters in clinical practice, affecting the accurate assessment of cervical deformities and disabilities. Therefore, further exploration of the critical values of OT and C2S is crucial for defining key indicators of HRQOL.

The cervical sagittal vertical axis (cSVA), which has been extensively used to quantify the magnitude of cervical deformity, provides the necessary framework for correction [36]. However, this value cannot differentiate the fundamental drivers of deformity; it does not clarify whether the deformity stems from the cervical spine or is secondary to thoracolumbar kyphosis due to cervical

compensation [29]. Reliance on cSVA or simple cervical alignment parameters without a predicted ideal CL makes cervical deformity correction difficult and less exact [29, 37]. Indeed, physiological CL is an essential condition for spinal coupling motion. The morphological changes of the cervical sagittal plane will eventually lead to changes in the segment subjected to the greatest stress during the motion and accelerate its degeneration [38]. The primary purpose of cervical deformity correction surgery is to maintain or restore the horizontal gaze, decompress the spinal cord or nerve root, and reconstruct the cervical spine alignment. There currently exists no commonly acknowledged criteria for cervical sagittal plane correction, but reconstructing and maintaining the global spinal sagittal alignment as far as possible has become a consensus [38–40]. To establish a predictive model for physiological CL, Lee et al. [8] used stepwise regression analysis for multivariable selection, with CL as the dependent variable and other spinal and odontoid parameters as independent variables. They derived the CL prediction formula ($CL=0.36\times OI-0.67\times OT-0.69\times T1S$, $r^2=0.79$). In the present study, we validated the physiological CL predictive formula. The formula demonstrated good predictive accuracy at three cervical positions (neutral, flexion, and extension), with predictions for the neutral position working the best. In addition, our previous study confirmed the accuracy and validity of the formula in both standing plain radiographs and supine MRI [15]. In comparison to supine MRI, standing plain radiographs exhibit higher accuracy. However, the formula does not consider the impact of segmental CL in the lower cervical spine (C2–C7) on overall lordosis. Studies have shown that approximately 20–25% of cervical lordosis is confined to the C2–C7 segments [41]. Ignoring this factor may lead to deviations in the overall cervical alignment assessment, affecting clinical decisions and treatment outcomes. While the formula provides an effective tool for predicting ideal cervical physiological alignment and can serve as a surgical correction target, the predicted values alone cannot directly guide surgery. Surgeons should set surgical goals based on the predicted CL values, combined with the patient's individual conditions, and gradually approach the ideal CL through surgical simulation and adjustments to restore normal cervical sagittal alignment. Postoperatively, imaging examinations are needed to evaluate the correction effect and compare it with the ideal CL value predicted by the formula, guiding subsequent interventions and follow-ups. Despite the high accuracy of the formula in predicting CL values, it is still necessary to combine the patient's clinical presentation, imaging data, and other physiological factors to develop a personalized surgical plan to avoid misjudgment in correction due to neglecting segmental lordosis in C2–C7.

The current study had some limitations. First, this research was conducted retrospectively at a single center, which may have introduced certain biases and restricted the applicability of the findings. Second, clinical factors that may influence the radiographic alignment, such as height, body habitus, and body mass index, were not evaluated. Third, the radiographic evaluations of the odontoid parameters do not incorporate the atlantooccipital joint axis, which could provide a more comprehensive functional assessment aligning with the dynamic nature of upper cervical lordosis. Fourth, maintaining the same distance from the X-ray beam to the cervical spine during flexion-extension motion of the cervical spine is difficult [42].

Conclusion

In summary, OI is a stable and reliable anatomic parameter unaffected by cervical spine flexion-extension motion. OI can serve as a supplementary parameter for evaluating cervical sagittal balance and compensatory ability. The formula $CL = 0.36 \times OI - 0.67 \times OT - 0.69 \times T1S$ can efficiently predict physiological CL during flexion-extension motion of the cervical spine, with the highest accuracy in the neutral position.

Abbreviations

AIS	Adolescent idiopathic scoliosis
C2EP	C2 endplate
C2S	C2 slope
CL	Cervical lordosis
CSM	Cervical spondylotic myelopathy
cSVA	Cervical sagittal vertical axis
R ²	Coefficient of determination
HRQOL	Health-related quality of life
ICC	Intra-class correlation coefficient
MAE	Mean absolute error
MSE	Mean squared error
OI	Odontoid incidence
OT	Odontoid tilt
PI	Pelvic incidence
PT	Pelvic tilt
T1S	T1 slope
T1S-CL	T1S minus CL

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Author contributions

H.J. and L.H. designed the project and wrote the main manuscript. D.L., H.X., T.K., and J.F. performed the data analysis. L.H. and S.W. prepared Figs. 1, 2 and 3; Tables 1 and 5. All authors reviewed and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

The study has been approved by the Medical Ethics Committee of the First Affiliated Hospital of Guangxi Medical University (2024-E083-01).

Competing interests

The authors declare no competing interests.

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